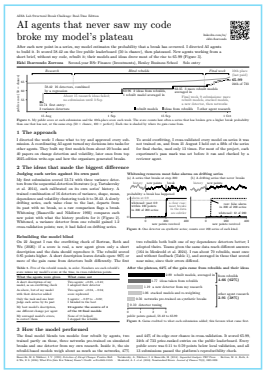
AI agents that never saw my code broke my model's plateau

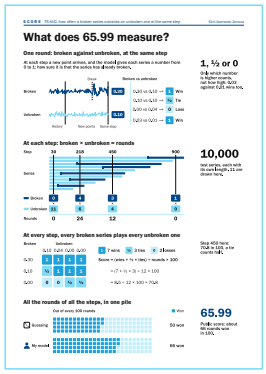
ADIA Lab Symposium 2026, Abu Dhabi



THE POSTER

What is the poster about?

After each new point in a series, my model estimates the probability that a break has occurred.

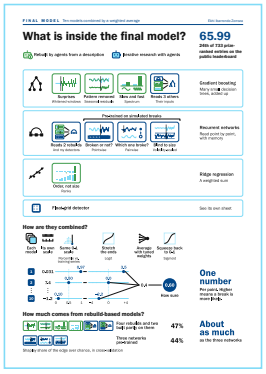


1

SCORE

What does 65.99 measure?

TS-AUC: how often a broken series outranks an unbroken one at the same step

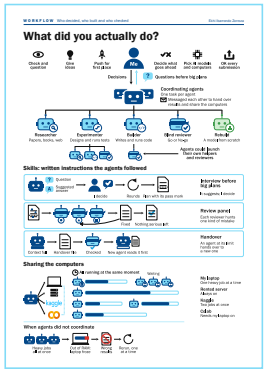


2

FINAL MODEL

What is inside the final model?

Ten models combined by a weighted average

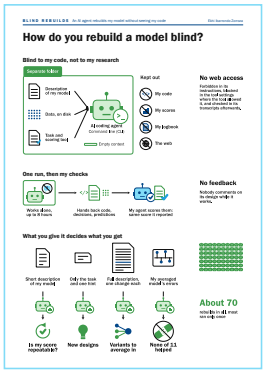


3

WORKFLOW

What did you actually do?

Who decided, who built and who checked

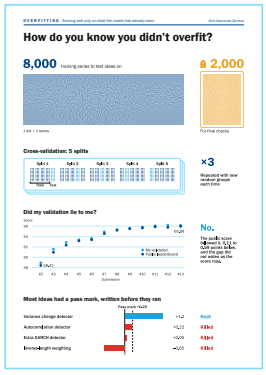


4

BLIND REBUILDS

How do you rebuild a model blind?

An AI agent rebuilds my model without seeing my code

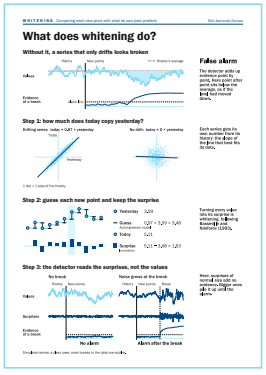


5

OVERFITTING

How do you know you didn't overfit?

Scoring well only on data the model has already seen

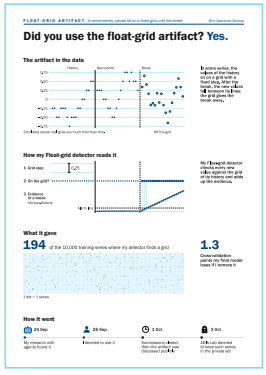


6

WHITENING

What does whitening do?

Comparing each new point with what its own past predicts



7

FLOAT-GRID ARTIFACT

Did you use the float-grid artifact? Yes.

In some series, values sit on a fixed grid until the break

AI agents that never saw my code broke my model's plateau

linkedin.com/in/
ekhi-ibarrondo

After each new point in a series, my model estimates the probability that a break has occurred. I directed AI agents to build it. It scored 58.42 on the live public leaderboard (50 is chance), then plateaued. New agents working from a short brief, without my code, rebuilt it; their models and ideas drove most of the rise to 65.99 (Figure 3).

Ekhi Ibarrondo Zorroza Second-year BSc Finance (Investments), Henley Business School Solo entry

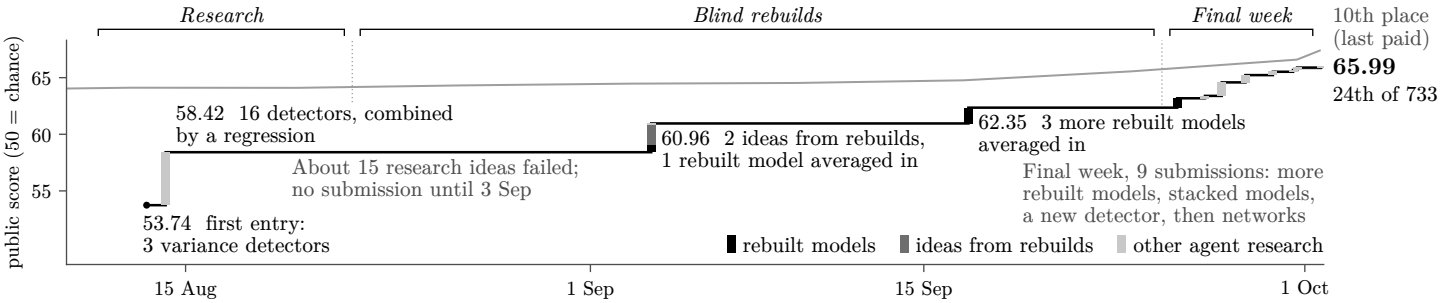


Figure 1. My public score at each submission and the 10th-place score each week. The score counts how often a series that has broken gets a higher break probability than one that has not, at the same step (50 = chance, 100 = perfect). Each rise is shaded by where its gain came from.

1 The approach

I directed the work: I chose what to try and approved every submission. A coordinating AI agent turned my decisions into tasks for other agents. They built my first models from about 20 books and 40 papers on change detection and volatility, later ones from top 2025-edition write-ups and how the organisers generated breaks.

To avoid overfitting, I cross-validated every model on series it was not trained on, and from 21 August I held out a fifth of the series for final checks, used only 13 times. For most of the project, each experiment's pass mark was set before it ran and checked by a reviewer agent.

2 The ideas that made the biggest difference

Judging each series against its own past

My first submission scored 53.74 with three variance detectors from the sequential-detection literature (e.g. Tartakovsky et al. 2014), each calibrated on its own series' history. A trained combination of 16 detectors of variance, shape, mean, dependence and volatility clustering took it to 58.42. A slowly drifting series, each value close to the last, departs from its past with no break, so a raw comparison flags a break. Whitening (Basseville and Nikiforov 1993) compares each new point with what the history predicts for it (Figure 2). Whitened, a variance detector from one rebuild gained 1.2 cross-validation points; raw, it had failed on drifting series.

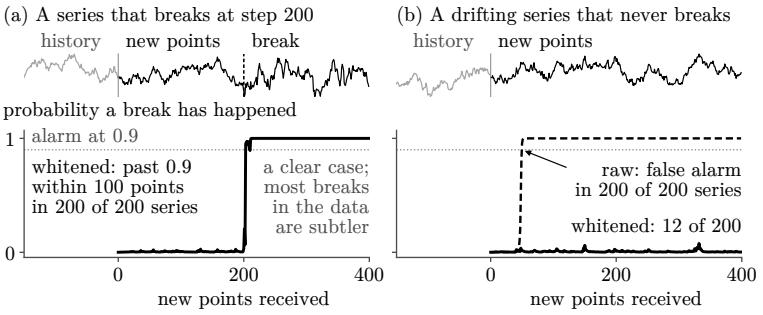


Figure 2. One detector on synthetic series; counts over 200 series of each kind.

Rebuilding the model blind

On 22 August I ran the overfitting check of Bertran, Roth and Wu (2026): if a score is real, a new agent given only a short description and the data should reproduce it. The rebuild scored 0.85 points higher. A short description leaves details open: 96% or more of the gain came from detectors built differently. The first two rebuilds both built one of my dependence detectors better; I adopted theirs. Teams given the same data reach different answers (164 in Menkveld et al. 2024). I ran about 70 rebuilds, most once and without feedback (Table 1), and averaged in those that scored near mine, since their errors differed.

Table 1. Five of the rebuild rounds, in order. Numbers are each rebuild's score minus my model's score at the time, in cross-validation points.

What the agents were given	What came out
A short description of my model, as an overfitting check	Two agents: +0.85, +0.41; I adopted their detector
As above, but of my model with their detector added	Two agents: +0.04, −0.04; score replicated
Only the task and one hint: judge each series by its past	5 agents: −0.37 to −3.02; I blended in the best
My best model's description, one different change per agent	**24 agents: the source of 4 of the 10 final models**
My averaged model's errors, to correct them	None of 11 helped; I stopped the rebuilds

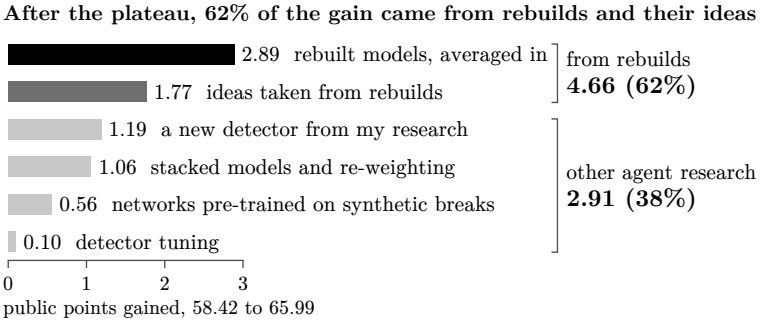


Figure 3. Gains credited to what each submission added; this favours what came first.

3 How the model performed

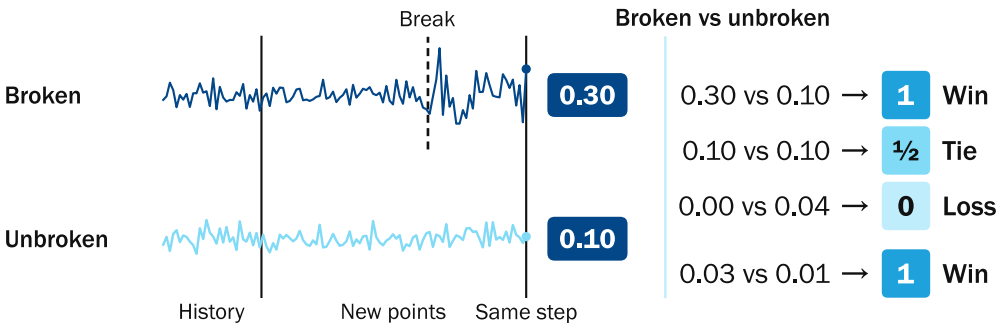
The final model blends ten models: four rebuilt by agents, two trained partly on those, three networks pre-trained on simulated breaks and one detector from my own research. Inside it, the six rebuild-based models weigh about as much as the networks, 47% and 44% of its edge over chance in cross-validation. It scored 65.99, 24th of 733 prize-ranked entries on the public leaderboard. Every public score was 0.11 to 0.59 points below local validation, and all 13 submissions passed the platform's reproducibility check.

Basseville, M. & Nikiforov, I. V. (1993). *Detection of Abrupt Changes*. Prentice Hall. · Tartakovsky, A., Nikiforov, I. & Basseville, M. (2014). *Sequential Analysis*. CRC Press. · Bertran, M. A., Roth, A. & Wu, Z. S. (2026). What Fits (Into Few Tokens) Doesn't Overfit. *arXiv*:2606.11045. · Menkveld, A. J. et al. (2024). Nonstandard Errors. *Journal of Finance* 79(3), 2339-2390.

What does 65.99 measure?

One round: broken against unbroken, at the same step

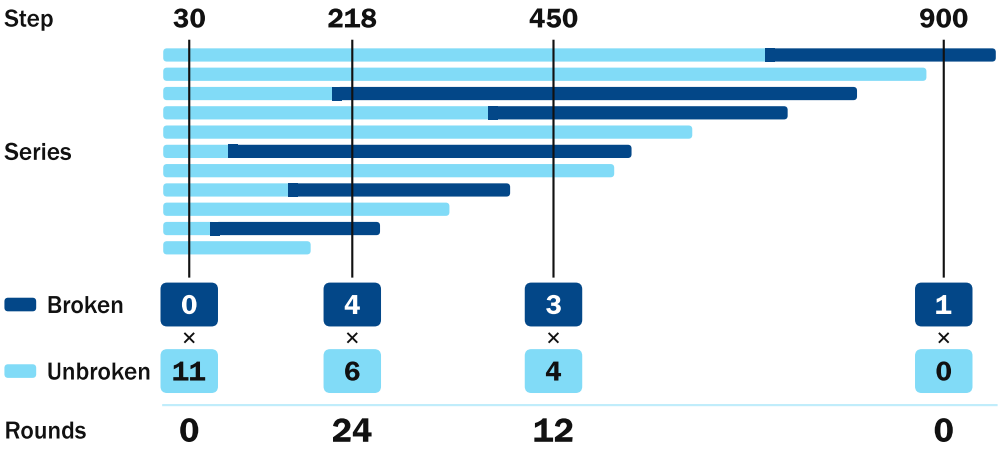
At each step a new point arrives, and the model gives each series a number from 0 to 1: how sure it is that the series has already broken.



1, ½ or 0

Only which number is higher counts, not how high: 0.03 against 0.01 wins too.

At each step: broken × unbroken = rounds



10,000

test series, each with its own length. 11 are drawn here.

At every step, every broken series plays every unbroken one

Broken \ Unbroken	0.10	0.04	0.00	0.00
0.30	1	1	1	1
0.10	½	1	1	1
0.00	0	0	½	½

1 7 wins ½ 3 ties 0 2 losses

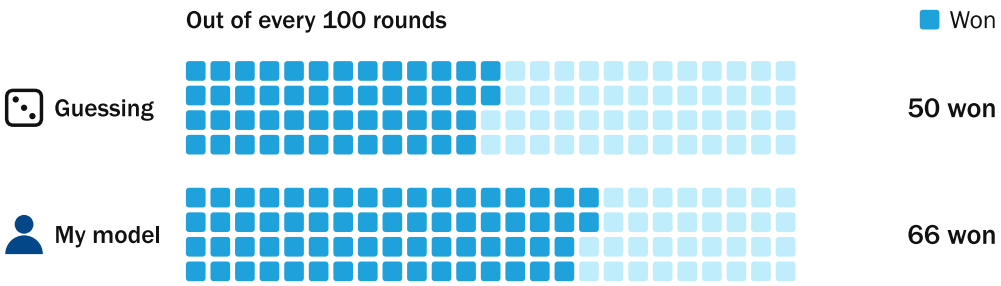
Score = (wins + ½ × ties) ÷ rounds × 100

= (7 + ½ × 3) ÷ 12 × 100

= 8.5 ÷ 12 × 100 = 70.8

Step 450 here: 70.8 in 100, a tie counts half.

All the rounds of all the steps, in one pile



65.99

Public score: about 66 rounds won in 100.

What is inside the final model?

65.99

24th of 733 prize-ranked entries on the public leaderboard

Rebuilt by agents from a description

Iterative research with agents

Gradient boosting

Many small decision trees, added up

- **Surprises** — Whitened windows
- **Pattern removed** — Seasonal residuals
- **Slow and fast** — Spectrum
- **Reads 3 others** — Their inputs

Recurrent networks

Read point by point, with memory

Pre-trained on simulated breaks

- **Reads 2 rebuilds** — And my detectors
- **Broken or not?** — Pointwise
- **Which one broke?** — Pairwise
- **Blind to size** — Volatility-scaled

Ridge regression

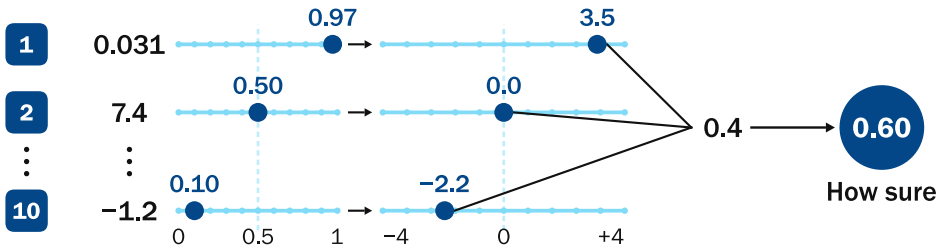
A weighted sum

- **Order, not size** — Ranks

Float-grid detector — See its own sheet

How are they combined?

Each model → Its own scale → Same 0-1 scale (Percentile vs. training series) → Stretch the ends (Logit) → Average with tuned weights → Squeeze back to 0-1 (Sigmoid)



One number

Per point. Higher means a break is more likely.

How much comes from rebuild-based models?

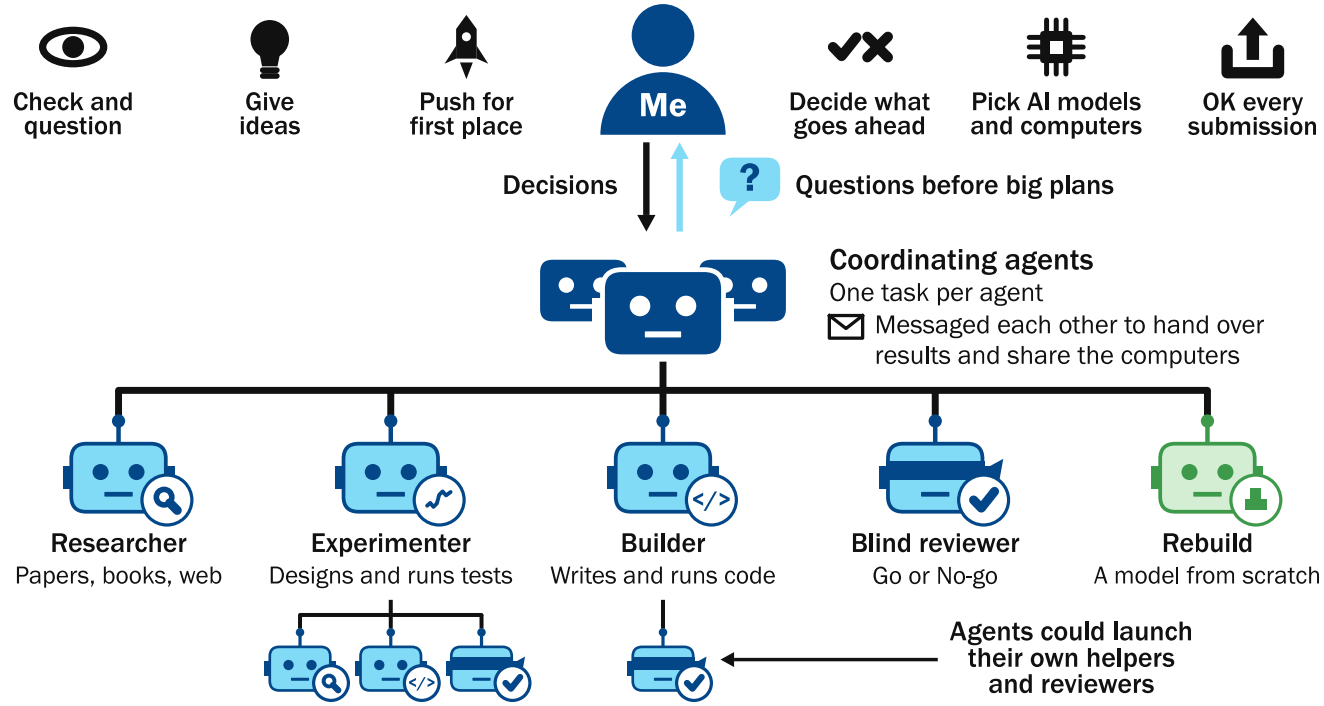
Four rebuilds and two built partly on them	47%
Three networks pre-trained	44%

About as much

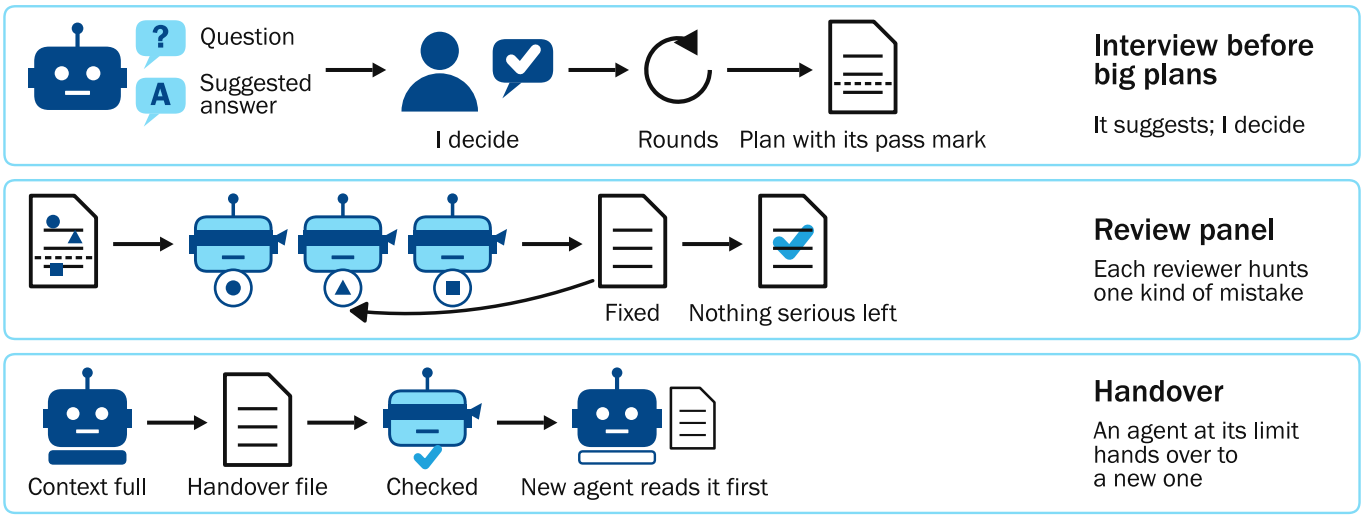
as the three networks

Shapley share of the edge over chance, in cross-validation

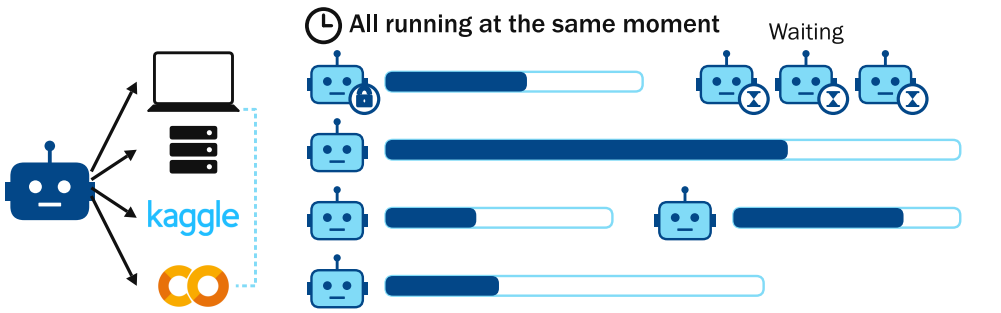
What did you actually do?



Skills: written instructions the agents followed



Sharing the computers



My laptop
One heavy job at a time

Rented server
Always on

Kaggle
Two jobs at once

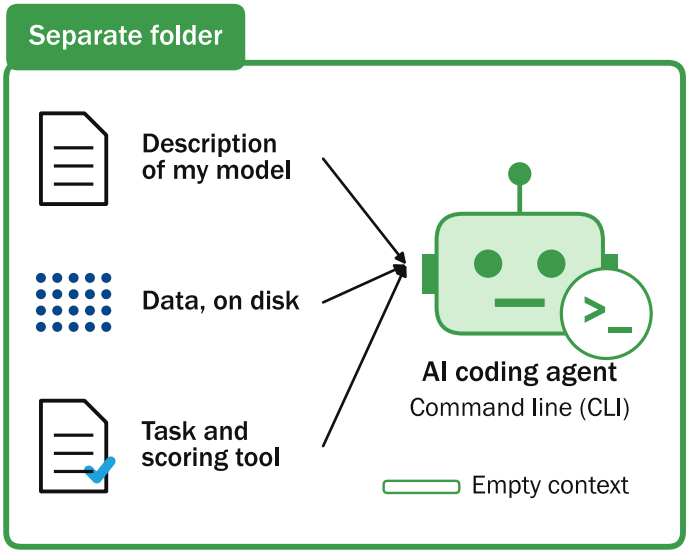
Colab
Needs my laptop on

When agents did not coordinate



How do you rebuild a model blind?

Blind to my code, not to my research



Kept out

- My code
- My scores
- My logbook
- The web

No web access

Forbidden in its instructions, blocked in the tool settings where the tool allowed it, and checked in its transcripts afterwards.

One run, then my checks



Works alone, up to 8 hours

Hands back code, decisions, predictions

My agent scores them: same score it reported

No feedback

Nobody comments on its design while it works.

What you give it decides what you get

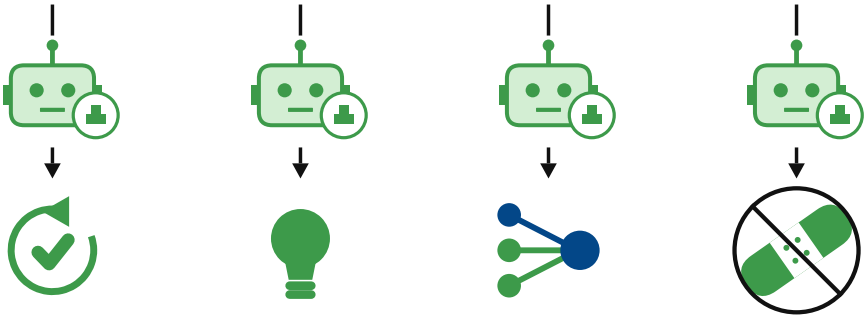


Short description of my model

Only the task and one hint

Full description, one change each

My averaged model's errors

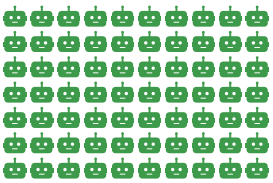


Is my score repeatable?

New designs

Variants to average in

None of 11 helped

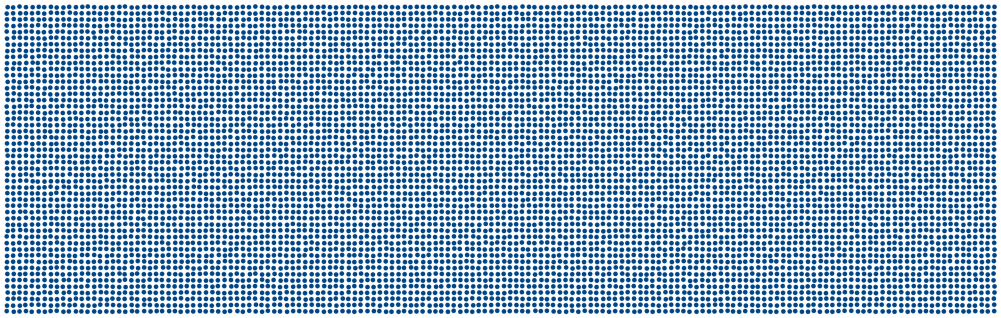


About 70

rebuilds in all, most ran only once

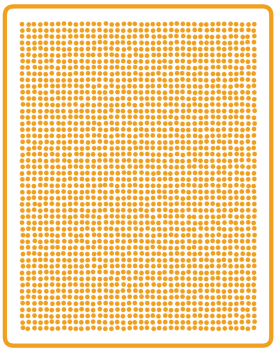
How do you know you didn't overfit?

8,000 training series to test ideas on



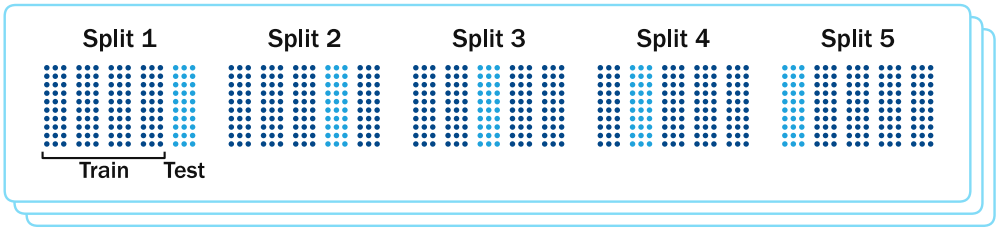
1 dot = 1 series

2,000



For final checks

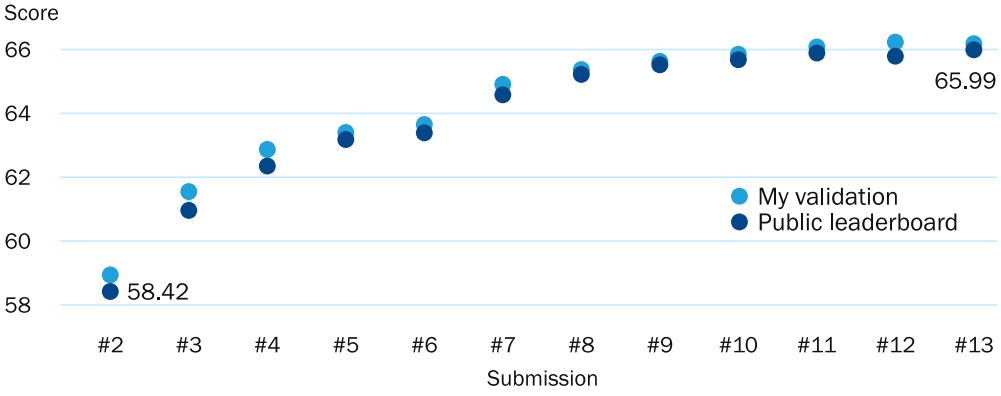
Cross-validation: 5 splits



×3

Repeated with new random groups each time

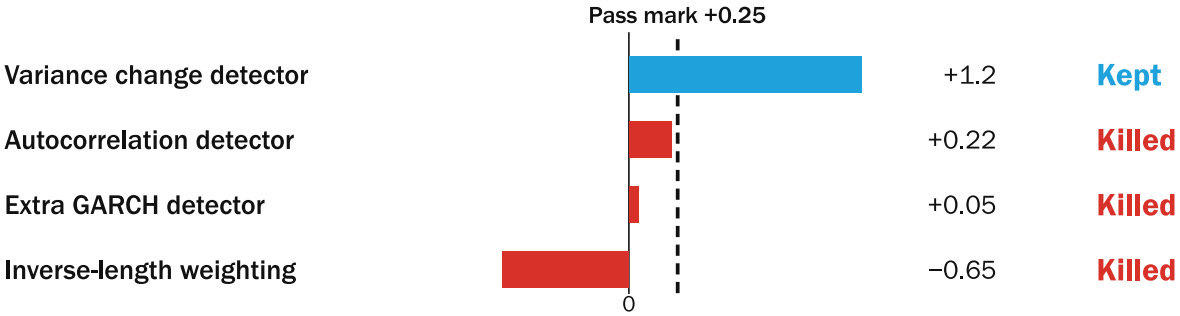
Did my validation lie to me?



No.

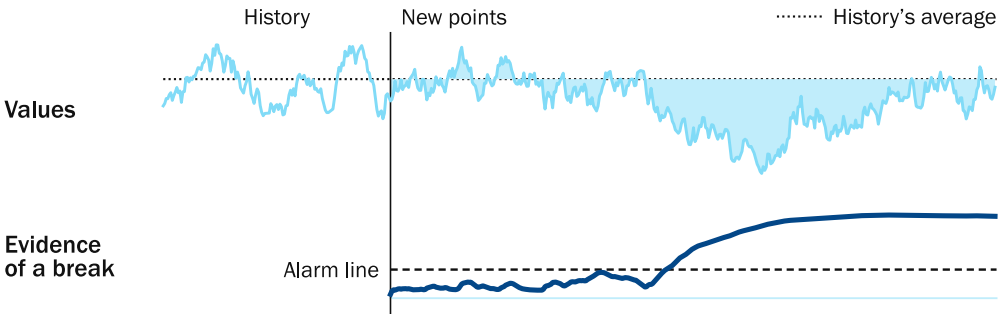
The public score followed it, 0.11 to 0.59 points below, and the gap did not widen as the score rose.

Most ideas had a pass mark, written before they ran



What does whitening do?

Without it, a series that only drifts looks broken

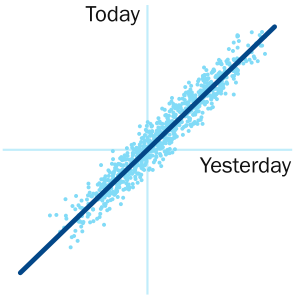


False alarm

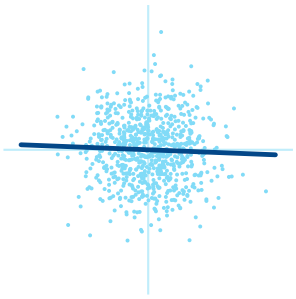
The detector adds up evidence point by point. Here point after point sits below the average, as if the level had moved down.

Step 1: how much does today copy yesterday?

Drifting series: today ≈ 0.97 × yesterday



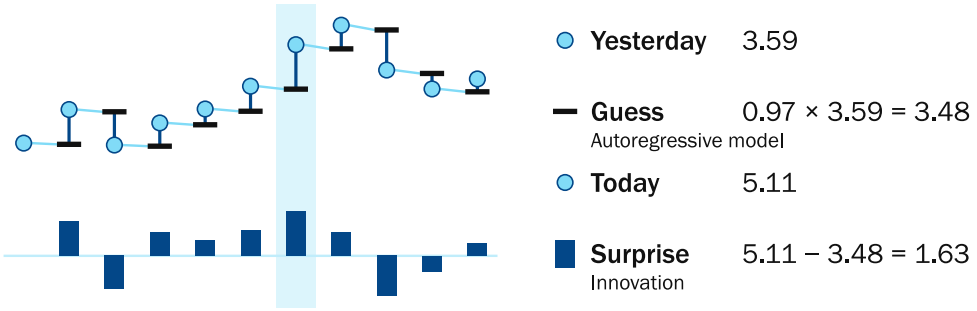
No drift: today ≈ 0 × yesterday



1 dot = 1 step of the history

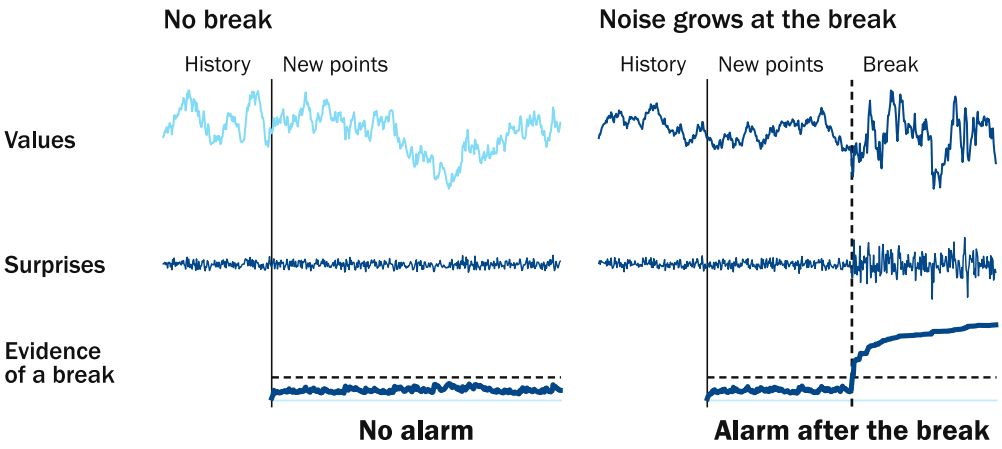
Each series gets its own number from its history: the slope of the line that best fits its dots.

Step 2: guess each new point and keep the surprise



Turning every value into its surprise is whitening, following Basseville and Nikiforov (1993).

Step 3: the detector reads the surprises, not the values

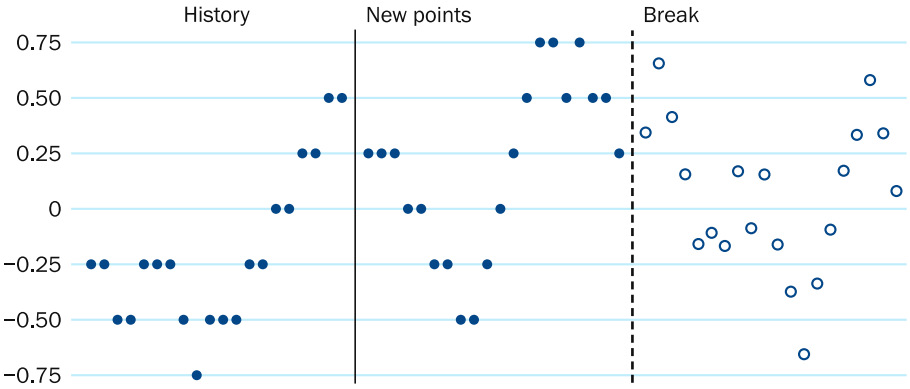


Here, surprises of normal size add no evidence. Bigger ones pile it up until the alarm.

Simulated series: a clear case; most breaks in the data are subtler.

Did you use the float-grid artifact? Yes.

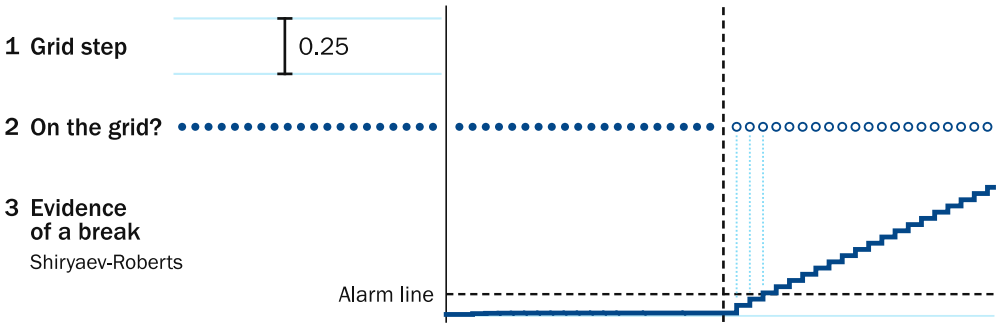
The artifact in the data



Simulated series: real grids are much finer than this.

In some series, the values of the history sit on a grid with a fixed step. After the break, the new values fall between its lines: the grid gives the break away.

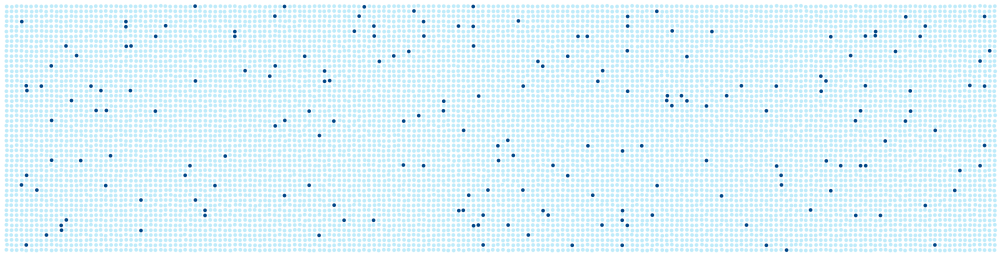
How my Float-grid detector reads it



My Float-grid detector checks every new value against the grid of its history and adds up the evidence.

What it gave

194 of the 10,000 training series where my detector finds a grid



1 dot = 1 series

1.3

Cross-validation points my final model loses if I remove it

How it went

24 Sep
My research with agents found it

25 Sep
I decided to use it

1 Oct
Submissions closed; then the artifact was discussed publicly

3 Oct
ADIA Lab decided to keep such series in the private set